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MODEL SELECTION AND VALIDATION REPORT ( fishing_teve model )
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## OBSERVATIONS:1755

## SANITY: warning − sanity check failed for the following models:
spatiotemporal_random_fields_iid,
 spatiotemporal_random_fields_iid_with_anisotropy,
 spatiotemporal_random_fields_ar1,
 spatiotemporal_random_fields_ar1_with_anisotropy

## SELECTED RANDOM TERM: spatial_random_fields

## MODEL SUMMARY ( fishing_teve model )

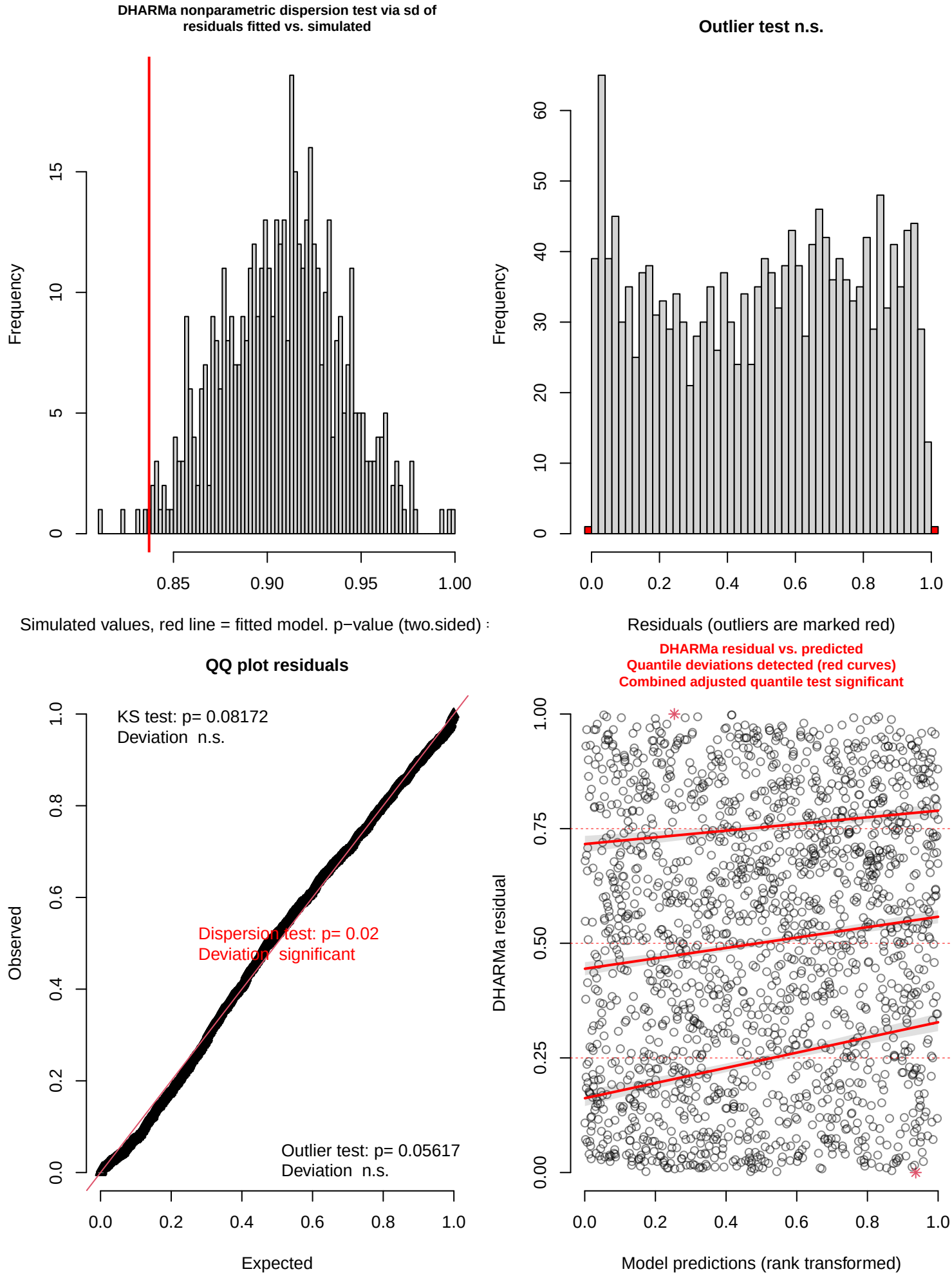
Spatial model fit by REML ['sdmTMB']
Formula: fixed_formula
Family: gaussian(link = 'identity')

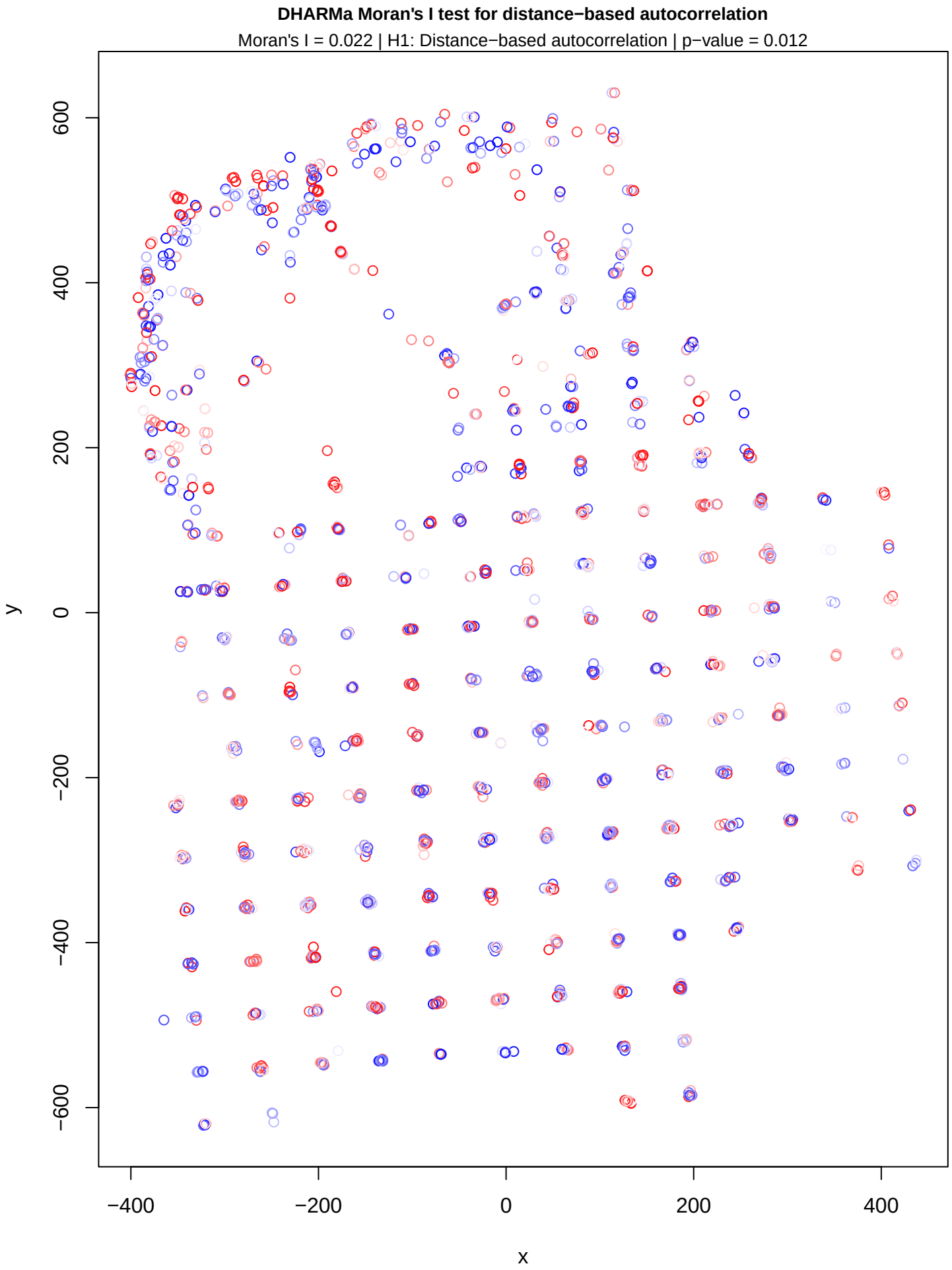
Conditional model:
                coef.est coef.se
(Intercept)        −0.02    0.11
log_sum_fishing    −0.03    0.03
sbt_mean            0.09    0.07
sbt_sd             −0.07    0.04
sst_sd              0.04    0.03
sic_mean           −0.15    0.05
log_chla_mean      −0.01    0.03
depth              −0.01    0.04

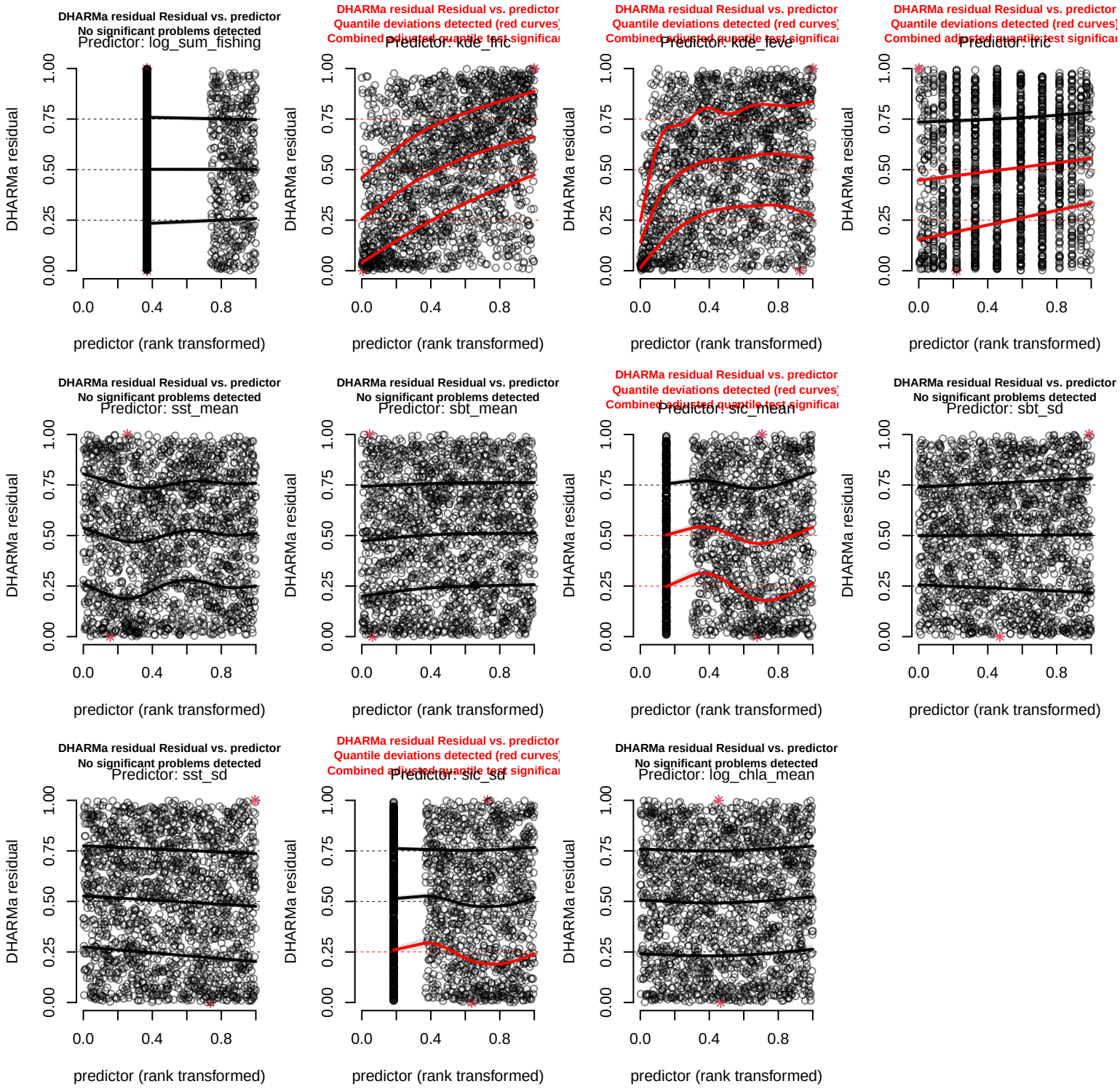
Dispersion parameter: 0.94
Matérn range: 254.75
Spatial SD: 0.37
REML criterion at convergence: 2455.013

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                  df      AIC delta_AIC
spatial_random_fields                             11 4932.025  3.929240
spatial_random_fields_with_anisotropy             13 4935.954 28.965007
spatiotemporal_random_fields_ar1                  12 4964.919  3.952871
spatiotemporal_random_fields_ar1_with_anisotropy 14 4968.872 33.959749
error_only                                         9 5002.832  4.000000
spatiotemporal_random_fields_iid                  11 5006.832  4.000000
spatiotemporal_random_fields_iid_with_anisotropy 13 5010.832        NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided) :
Outlier test n.s.
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.08172
Deviation n.s.
Dispersion test: p= 0.02
Deviation significant
Outlier test: p= 0.05617
Deviation n.s.
Observed
Expected
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.022 | H1: Distance−based autocorrelation | p−value = 0.012
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: log_sum_fishing
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Predictor: kde_fric
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Predictor: kde_feve
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Predictor: tric
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sst_mean
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sbt_mean
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Predictor: sic_mean
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sbt_sd
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: sst_sd
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Predictor: sic_sd
DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8